

Table 4: Performance evaluation of KIVI on various models across a range of benchmarks in LongBench. We highlight the average performance of our method. **More similar results on Mistral-7B-v0.2 and LongChat-7b-v1.5 can be found in Table 10 and Table 9**

Model		Qasper	QMSum	MultiNews	TREC	TriviaQA	SAMSum	LCC	RepoBench-P	Average
Llama2-7B	16bit	9.52	21.28	3.51	66.00	87.72	41.69	66.66	59.82	44.52
	KIVI-4	9.28	21.42	3.88	66.00	87.72	41.82	66.80	59.83	44.59
	KIVI-2	9.31	20.50	1.14	66.00	87.42	42.71	66.88	60.23	44.27
Llama2-13B	16bit	9.32	21.38	3.71	70.00	87.87	43.55	66.61	56.42	44.85
	KIVI-4	9.16	20.86	3.21	69.00	86.97	44.26	65.30	57.08	44.48
	KIVI-2	8.58	20.69	6.19	69.50	87.78	44.30	65.08	55.46	44.69
Llama2-7B-Chat	16bit	19.65	20.54	26.36	63.00	84.28	41.12	59.75	52.93	45.95
	KIVI-4	19.62	20.70	25.49	63.00	84.13	40.87	59.27	53.56	45.83
	KIVI-2	19.32	20.46	25.48	63.00	84.84	40.60	58.71	52.97	45.67
Llama2-13B-Chat	16bit	24.18	20.37	25.69	67.50	86.90	42.18	50.23	50.64	45.96
	KIVI-4	23.00	20.36	26.06	67.50	87.20	42.04	52.55	52.77	46.44
	KIVI-2	23.59	20.76	25.25	67.5	87.17	41.56	49.93	48.45	45.52
Falcon-7B	16bit	1.48	2.35	11.09	13.00	5.84	2.44	23.86	9.69	8.71
	KIVI-4	1.04	2.41	11.98	13.00	5.84	2.36	23.72	9.92	8.78
	KIVI-2	1.98	3.61	6.78	10.00	6.24	2.73	22.18	10.12	7.95
Mistral-7B	16bit	8.12	19.98	19.99	67.50	89.80	41.69	66.59	58.99	46.58
	KIVI-4	7.89	20.06	20.58	67.50	89.80	41.56	66.45	58.62	46.56
	KIVI-2	6.92	19.71	17.92	66.50	89.63	41.66	65.52	58.99	45.85

Table 5: Ablation study of KIVI by changing group size G and residual length R .

Model	Group Size	GSM8K
Llama2-13B	32	20.77
	64	21.00
	128	17.29
Model	Residual Length	GSM8K
Llama2-13B	32	20.62
	64	19.86
	96	20.55
	128	20.77

One essential recipe of KIVI is the per-channel quantization scheme designed based on the observation made in Section 3.2 and Figure 2. ATOM (Zhao et al., 2024) also indicates that key cache exhibits more outliers compared to the value cache. KIVI provides further extensive analysis and leverages this observation to implement per-channel quantization. A similar observation and approach has been independently discovered and developed in the concurrent work KVQuant (Hooper et al., 2024).

vLLM (Kwon et al., 2023) and S3 (Jin et al., 2023) are system-level works, which include memory management through the use of PagedAttention or memory usage prediction. They can lower the memory requirements of KV cache and simultaneously increase model throughput. This research direction is orthogonal to our work, since system-level optimizations can also be applied upon our algorithm.

Several other works also consider compressing KV cache by evicting tokens. H2O (Zhang et al., 2023) retains only a small portion of tokens that contribute significantly to the attention scores. Similarly, Scissorhands (Liu et al., 2024) exploit the persistence of the importance hypothesis in KV cache sparsification. StreamingLLM (Xiao et al., 2023b) is based on the observation of “attention sink” and maintains only a few initial tokens to preserve performance. Unlike these works, our KIVI retains all input tokens and compresses them into lower precision. This line of work is orthogonal to ours, as they can also be combined together.

6. Conclusion and Future Work

In this paper, we systematically analyze KV cache element distribution in popular LLMs. We conclude that key cache should be quantized per-channel and value cache should be quantized per token. Based on these observations, we propose KIVI, a plug-and-play 2bit KV cache quantization algorithm without the need for any tuning. In real LLM workload, KIVI allows up to $4\times$ larger batch sizes and $3.47\times$ throughput. In the future, we will further optimize the implementation to reduce the overhead of quantization process during the prefill and decoding phase.

Acknowledgments

The authors thank the anonymous reviewers for their helpful comments. Dr. Beidi Chen is supported by a research gift from Moffett.AI. Dr. Vladimir Braverman is partially

supported by the Ministry of Trade, Industry and Energy (MOTIE) and Korea Institute for Advancement of Technology (KIAT) through the International Cooperative R&D program, the Naval Research (ONR) grant N00014-23-1-2737, and NSF CNS 2333887 award. Dr. Xia Hu is supported by NSF grants NSF IIS-2224843. The views and conclusions contained in this paper are those of the authors and should not be interpreted as representing any funding agencies.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- ShareGPT Team. <https://sharegpt.com/>, 2023.
- Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints. *arXiv preprint arXiv:2305.13245*, 2023.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. Longbench: A bilingual, multitask benchmark for long context understanding. *arXiv preprint arXiv:2308.14508*, 2023.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- Yu-Neng Chuang, Songchen Li, Jiayi Yuan, Guanchu Wang, Kwei-Herng Lai, Leisheng Yu, Sirui Ding, Chia-Yuan Chang, Qiaoyu Tan, Daochen Zha, et al. Understanding different design choices in training large time series models. *arXiv preprint arXiv:2406.14045*, 2024.
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. Llm.int8(): 8-bit matrix multiplication for transformers at scale. *arXiv preprint arXiv:2208.07339*, 2022.
- Elias Frantar, Saleh Ashkboos, Torsten Hoeftler, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, et al. A framework for few-shot language model evaluation. *Version v0.0.1. Sept*, page 8, 2021.
- Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. Kvquant: Towards 10 million context length llm inference with kv cache quantization. *arXiv preprint arXiv:2401.18079*, 2024.
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Yunho Jin, Chun-Feng Wu, David Brooks, and Gu-Yeon Wei. S3: Increasing gpu utilization during generative inference for higher throughput. *arXiv preprint arXiv:2306.06000*, 2023.
- Sehoon Kim, Coleman Hooper, Amir Gholami, Zhen Dong, Xiuyu Li, Sheng Shen, Michael W Mahoney, and Kurt Keutzer. Squeezellm: Dense-and-sparse quantization. *arXiv preprint arXiv:2306.07629*, 2023.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626, 2023.
- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Xingyu Dang, and Song Han. Awq: Activation-aware weight quantization for llm compression and acceleration. *arXiv preprint arXiv:2306.00978*, 2023.
- Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhao Xu, Anastasios Kyrillidis, and Anshumali Shrivastava. Scissorhands: Exploiting the persistence of importance hypothesis for llm kv cache compression at test time. *Advances in Neural Information Processing Systems*, 36, 2024.
- Amirkeivan Mohtashami and Martin Jaggi. Landmark attention: Random-access infinite context length for transformers. *arXiv preprint arXiv:2305.16300*, 2023.
- Guilherme Penedo, Quentin Malartic, Daniel Hesslow, Ruxandra Cojocar, Alessandro Cappelli, Hamza Alobeidli, Baptiste Pannier, Ebtesam Almazrouei, and Julien Launay. The refinedweb dataset for falcon llm: outperforming curated corpora with web data, and web data only. *arXiv preprint arXiv:2306.01116*, 2023.

- Reiner Pope, Sholto Douglas, Aakanksha Chowdhery, Jacob Devlin, James Bradbury, Jonathan Heek, Kefan Xiao, Shivani Agrawal, and Jeff Dean. Efficiently scaling transformer inference. *Proceedings of Machine Learning and Systems*, 5, 2023.
- Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy Lillicrap, Jean-baptiste Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Firat, Julian Schrittwieser, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Noam Shazeer. Fast transformer decoding: One write-head is all you need. *arXiv preprint arXiv:1911.02150*, 2019.
- Ying Sheng, Lianmin Zheng, Binhang Yuan, Zhuohan Li, Max Ryabinin, Beidi Chen, Percy Liang, Christopher Ré, Ion Stoica, and Ce Zhang. Flexgen: High-throughput generative inference of large language models with a single gpu. In *International Conference on Machine Learning*, pages 31094–31116. PMLR, 2023.
- Ross Taylor, Marcin Kardas, Guillem Cucurull, Thomas Scialom, Anthony Hartshorn, Elvis Saravia, Andrew Poulton, Viktor Kerkez, and Robert Stojnic. Galactica: A large language model for science. *arXiv preprint arXiv:2211.09085*, 2022.
- Yuandong Tian, Yiping Wang, Beidi Chen, and Simon Du. Scan and snap: Understanding training dynamics and token composition in 1-layer transformer. *arXiv preprint arXiv:2305.16380*, 2023.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023b.
- Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, pages 38087–38099. PMLR, 2023a.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023b.
- Zhaozhuo Xu, Zirui Liu, Beidi Chen, Yuxin Tang, Jue Wang, Kaixiong Zhou, Xia Hu, and Anshumali Shrivastava. Compress, then prompt: Improving accuracy-efficiency trade-off of llm inference with transferable prompt. *arXiv preprint arXiv:2305.11186*, 2023.
- Jiayi Yuan, Ruixiang Tang, Xiaoqian Jiang, and Xia Hu. Large language models for healthcare data augmentation: An example on patient-trial matching. In *AMIA Annual Symposium Proceedings*, volume 2023, page 1324. American Medical Informatics Association, 2023.
- Jiayi Yuan, Hongyi Liu, Shaochen Zhong, Yu-Neng Chuang, Songchen Li, Guanchu Wang, Duy Le, Hongye Jin, Vipin Chaudhary, Zhaozhuo Xu, Zirui Liu, and Xia Hu. Kv cache compression, but what must we give in return? a comprehensive benchmark of long context capable approaches. *arXiv preprint arXiv:2407.01527*, 2024.
- Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, et al. H₂o: Heavy-hitter oracle for efficient generative inference of large language models. *arXiv preprint arXiv:2306.14048*, 2023.
- Yilong Zhao, Chien-Yu Lin, Kan Zhu, Zihao Ye, Lequn Chen, Size Zheng, Luis Ceze, Arvind Krishnamurthy, Tianqi Chen, and Baris Kasikci. Atom: Low-bit quantization for efficient and accurate llm serving. *Proceedings of Machine Learning and Systems*, 6:196–209, 2024.

A. Detailed Implementations

In this section, we present the algorithm for KIVI as discussed in Section 3.3. Specifically, we provide the pseudocode for KIVI when calculating the attention output in the prefill and decoding phases.

Algorithm 1: The KIVI Prefill & Decoding Algorithm

parameter: group size G , residual length R

procedure Prefill:

```

Input:  $\mathbf{X} \in \mathbb{R}^{l_{\text{prompt}} \times d}$ 
 $\mathbf{X}_K = \mathbf{X} \mathbf{W}_K, \mathbf{X}_V = \mathbf{X} \mathbf{W}_V$ 
 $\mathbf{X}_{V_g} = \mathbf{X}_V[:, l_{\text{prompt}} - R:], \mathbf{X}_{V_r} = \mathbf{X}_V[l_{\text{prompt}} - R :]$ 
 $Q(\mathbf{X}_{V_g}) \leftarrow \text{GroupQuant}(\mathbf{X}_{V_g}, \text{dim}=\text{token}, \text{numGroup}=d//G)$ 
 $Q(\mathbf{X}_{K_g}), \mathbf{X}_{K_r} \leftarrow \text{KeyQuant}(\mathbf{X}_K)$ 
KV cache  $\leftarrow Q(\mathbf{X}_{K_g}), \mathbf{X}_{K_r}, Q(\mathbf{X}_{V_g}), \mathbf{X}_{V_r}$ 
return  $\mathbf{X}_K, \mathbf{X}_V$ 

```

end

procedure Decoding:

```

Input: KV cache,  $\mathbf{t} \in \mathbb{R}^{1 \times d}$ 
 $\mathbf{t}_Q = \mathbf{t} \mathbf{W}_Q, \mathbf{t}_K = \mathbf{t} \mathbf{W}_K, \mathbf{t}_V = \mathbf{t} \mathbf{W}_V$ 
 $Q(\mathbf{X}_{K_g}), \mathbf{X}_{K_r}, Q(\mathbf{X}_{V_g}), \mathbf{X}_{V_r} \leftarrow \text{KV cache}$ 
 $\mathbf{X}_{K_r} \leftarrow \text{Concat}([\mathbf{X}_{K_r}, \mathbf{t}_K], \text{dim}=\text{token})$ 
 $\mathbf{X}_{V_r} \leftarrow \text{Concat}([\mathbf{X}_{V_r}, \mathbf{t}_V], \text{dim}=\text{token})$ 
if  $\text{len}(\mathbf{X}_{K_r}) = R$  then
     $Q(\mathbf{X}_{K_r})_{,-} \leftarrow \text{KeyQuant}(\mathbf{X}_{K_r})$ 
     $Q(\mathbf{X}_{K_g}) \leftarrow \text{Concat}([Q(\mathbf{X}_{K_g}), Q(\mathbf{X}_{K_r})], \text{dim}=\text{token})$ 
     $\mathbf{X}_{K_r} \leftarrow \text{empty tensor.}$ 
end
if  $\text{len}(\mathbf{X}_{V_r}) > R$  then
     $Q(\mathbf{X}_{V_r'}) \leftarrow \text{GroupQuant}(\mathbf{X}_{V_r}[:, -R], \text{dim}=\text{token}, \text{numGroup} = d//G)$ 
     $Q(\mathbf{X}_{V_g}) \leftarrow \text{Concat}([Q(\mathbf{X}_{V_g}), Q(\mathbf{X}_{V_r'})], \text{dim}=\text{token})$ 
     $\mathbf{X}_{V_r} \leftarrow \mathbf{X}_{V_r}[:, -R:]$ 
end
 $\mathbf{A} \leftarrow \text{Concat}([\mathbf{t}_Q Q(\mathbf{X}_{K_g})^\top, \mathbf{t}_Q \mathbf{X}_{K_r}^\top], \text{dim}=\text{token})$ 
 $\mathbf{A}_g = \text{Softmax}(\mathbf{A})[:, -R], \mathbf{A}_r = \text{Softmax}(\mathbf{A})[:, -R:]$ 
 $\mathbf{t}_O \leftarrow \mathbf{A}_g Q(\mathbf{X}_{V_g}) + \mathbf{A}_r \mathbf{X}_{V_r}$ 
KV cache  $\leftarrow Q(\mathbf{X}_{K_g}), \mathbf{X}_{K_r}, Q(\mathbf{X}_{V_g}), \mathbf{X}_{V_r}$ 
return  $\mathbf{t}_O$ 

```

end

function KeyQuant($\mathbf{X}_K \in \mathbb{R}^{l \times d}$):

```

 $r = l \% R,$ 
 $\mathbf{X}_{K_g} = \mathbf{X}_K[:, l - r], \mathbf{X}_{K_r} = \mathbf{X}_K[l - r :]$ 
 $Q(\mathbf{X}_{K_g}) \leftarrow \text{GroupQuant}(\mathbf{X}_{K_g}, \text{dim}=\text{channel}, \text{numGroup}=l//G)$ 
return  $Q(\mathbf{X}_{K_g}), \mathbf{X}_{K_r}$ 

```

end
